

```
In [1]: import pandas as pd
```

```
In [2]: movies = pd.read_csv(r'/Users/abhiramireddygali/Downloads/march(2021)
```

```
In [3]: movies
```

```
Out[3]:
```

	Film	Genre	Rotten Tomatoes Ratings %	Audience Ratings %	Budget (million \$)	Year of release
0	(500) Days of Summer	Comedy	87	81	8	2009
1	10,000 B.C.	Adventure	9	44	105	2008
2	12 Rounds	Action	30	52	20	2009
3	127 Hours	Adventure	93	84	18	2010
4	17 Again	Comedy	55	70	20	2009
...	...	...	...	...	...	...
554	Your Highness	Comedy	26	36	50	2011
555	Youth in Revolt	Comedy	68	52	18	2009
556	Zodiac	Thriller	89	73	65	2007
557	Zombieland	Action	90	87	24	2009
558	Zookeeper	Comedy	14	42	80	2011

559 rows × 6 columns

```
In [4]: type(movies)
```

```
Out[4]: pandas.core.frame.DataFrame
```

```
In [5]: movies.shape
```

```
Out[5]: (559, 6)
```

```
In [6]: import numpy
```

```
In [7]: movies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 559 entries, 0 to 558
Data columns (total 6 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Film                                  559 non-null    object
1   Genre                                559 non-null    object
2   Rotten Tomatoes Ratings %           559 non-null    int64
3   Audience Ratings %                  559 non-null    int64
4   Budget (million $)                  559 non-null    int64
5   Year of release                      559 non-null    int64
dtypes: int64(4), object(2)
memory usage: 26.3+ KB
```

```
In [9]: movies.columns
```

```
Out[9]: Index(['Film', 'Genre', 'Rotten Tomatoes Ratings %', 'Audience Rat
ings %',
              'Budget (million $)', 'Year of release'],
              dtype='object')
```

```
In [10]: movies.head()
```

```
Out[10]:
```

	Film	Genre	Rotten Tomatoes Ratings %	Audience Ratings %	Budget (million \$)	Year of release
0	(500) Days of Summer	Comedy	87	81	8	2009
1	10,000 B.C.	Adventure	9	44	105	2008
2	12 Rounds	Action	30	52	20	2009
3	127 Hours	Adventure	93	84	18	2010
4	17 Again	Comedy	55	70	20	2009

```
In [11]: movies.tail()
```

```
Out[11]:
```

	Film	Genre	Rotten Tomatoes Ratings %	Audience Ratings %	Budget (million \$)	Year of release
554	Your Highness	Comedy	26	36	50	2011
555	Youth in Revolt	Comedy	68	52	18	2009
556	Zodiac	Thriller	89	73	65	2007
557	Zombieland	Action	90	87	24	2009
558	Zookeeper	Comedy	14	42	80	2011

```
In [12]: movies.describe()
```

Out[12]:

	Rotten Tomatoes Ratings %	Audience Ratings %	Budget (million \$)	Year of release
count	559.000000	559.000000	559.000000	559.000000
mean	47.309481	58.744186	50.236136	2009.152057
std	26.413091	16.826887	48.731817	1.362632
min	0.000000	0.000000	0.000000	2007.000000
25%	25.000000	47.000000	20.000000	2008.000000
50%	46.000000	58.000000	35.000000	2009.000000
75%	70.000000	72.000000	65.000000	2010.000000
max	97.000000	96.000000	300.000000	2011.000000

In [13]: `movies.columns = ['Film', 'Genre', 'CriticalRating', 'AudienceRating',`In [14]: `movies.columns`Out[14]: `Index(['Film', 'Genre', 'CriticalRating', 'AudienceRating', 'BudgetMillions',
 'Year'],
 dtype='object')`In [16]: `movies.head(1)`

Out[16]:

	Film	Genre	CriticalRating	AudienceRating	BudgetMillions	Year
	(500)					
0	Days of Summer	Comedy	87	81	8	2009

In [17]: `movies.Film = movies.Film.astype('category')`In [19]: `movies.Film`

Out[19]:

```

0      (500) Days of Summer
1      10,000 B.C.
2      12 Rounds
3      127 Hours
4      17 Again
...
554     Your Highness
555     Youth in Revolt
556     Zodiac
557     Zombieland
558     Zookeeper
Name: Film, Length: 559, dtype: category
Categories (559, object): ['(500) Days of Summer ', '10,000 B.C.',
'12 Rounds ', '127 Hours', ..., 'Youth in Revolt', 'Zodiac', 'Zombieland ', 'Zookeeper']

```

```
In [20]: movies.Genre = movies.Genre.astype('category')
         movies.Year = movies.Year.astype('category')
```

```
In [21]: movies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 559 entries, 0 to 558
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Film                  559 non-null   category
1   Genre                 559 non-null   category
2   CriticalRating        559 non-null   int64
3   AudienceRating        559 non-null   int64
4   BudgetMillions        559 non-null   int64
5   Year                  559 non-null   category
dtypes: category(3), int64(3)
memory usage: 36.5 KB
```

```
In [23]: movies.describe()
```

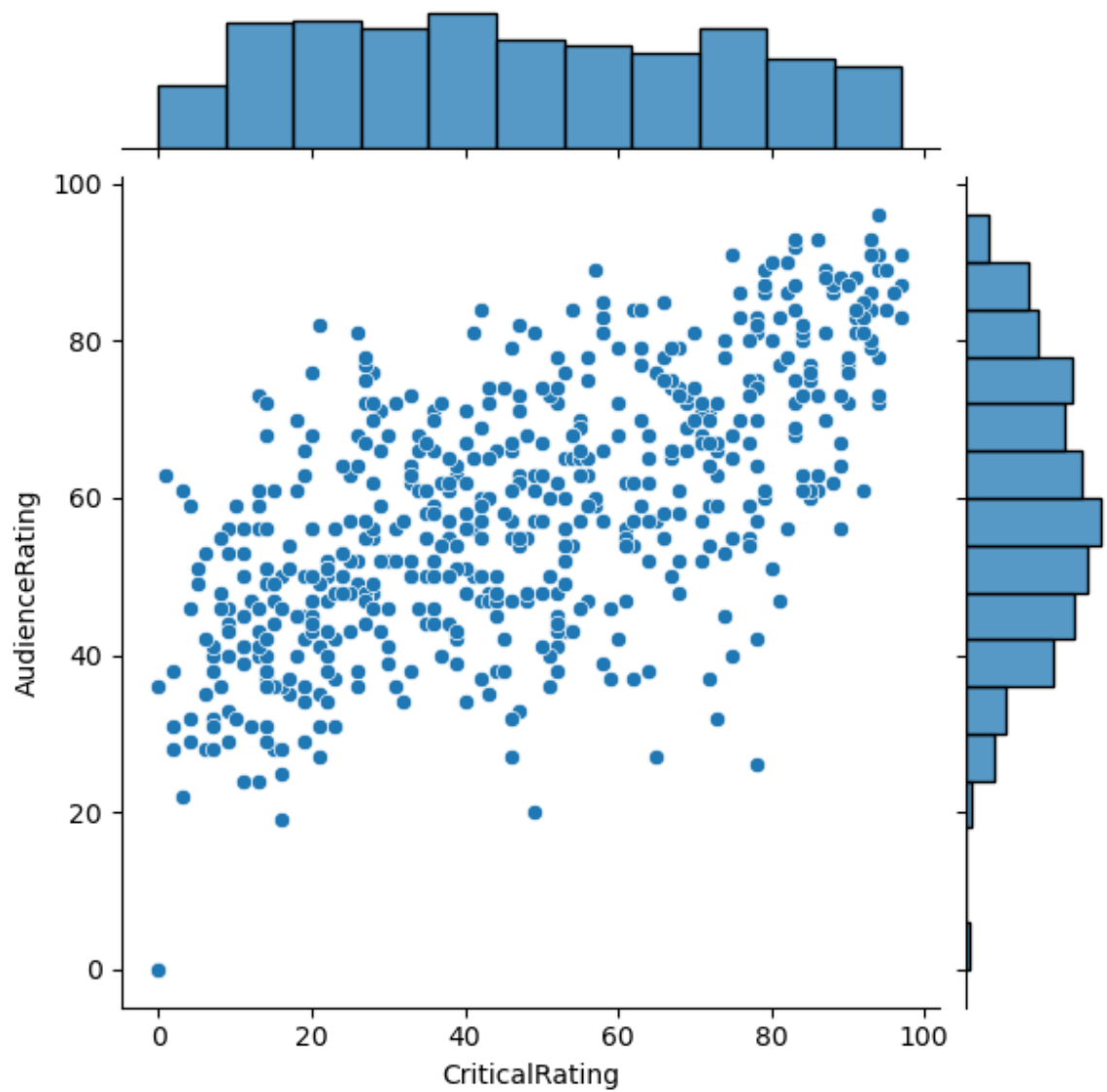
```
Out[23]:
```

	CriticalRating	AudienceRating	BudgetMillions
count	559.000000	559.000000	559.000000
mean	47.309481	58.744186	50.236136
std	26.413091	16.826887	48.731817
min	0.000000	0.000000	0.000000
25%	25.000000	47.000000	20.000000
50%	46.000000	58.000000	35.000000
75%	70.000000	72.000000	65.000000
max	97.000000	96.000000	300.000000

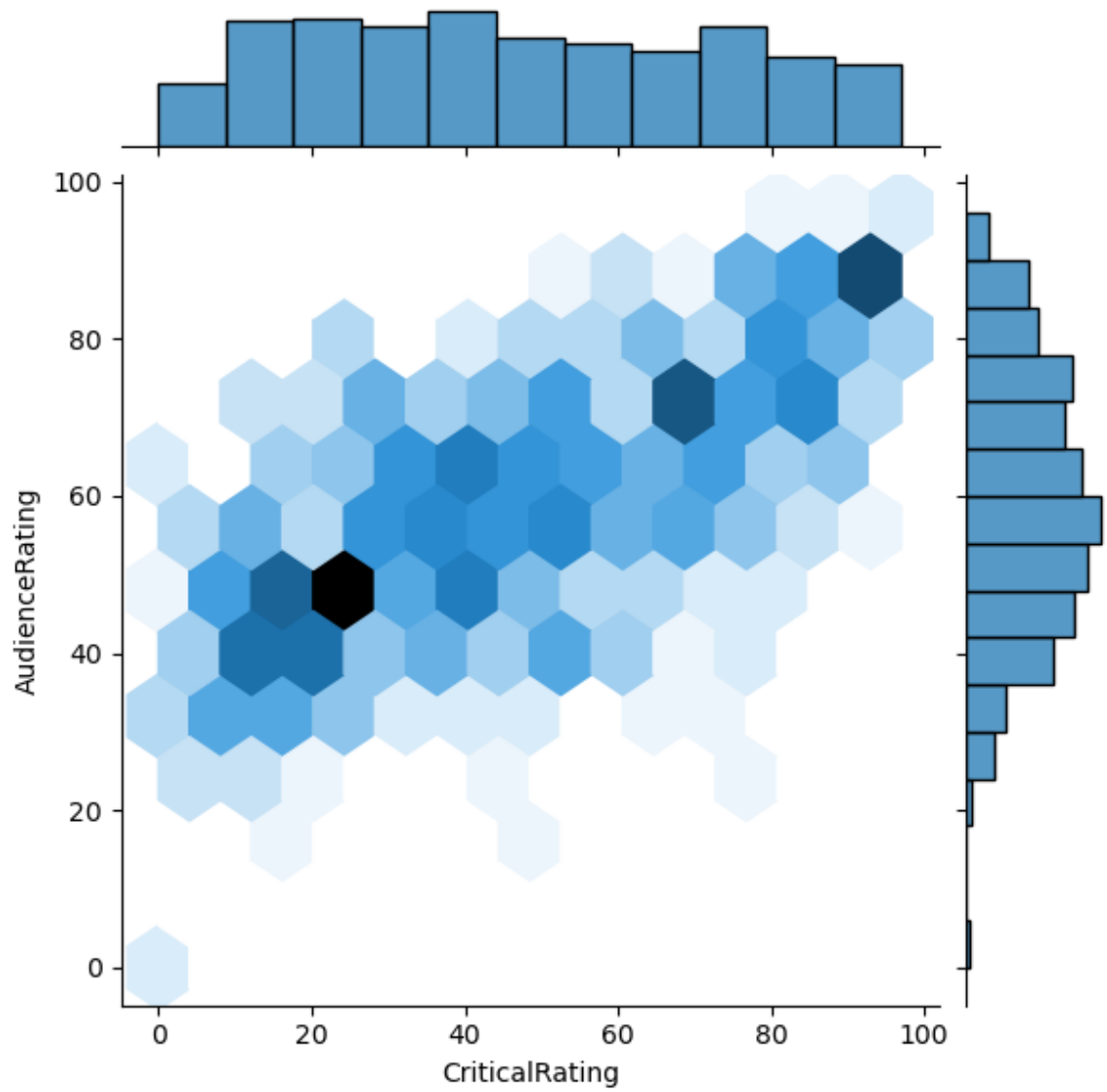
```
In [24]: from matplotlib import pyplot as plt
         import seaborn as sns

         %matplotlib inline
         import warnings
         warnings.filterwarnings('ignore')
```

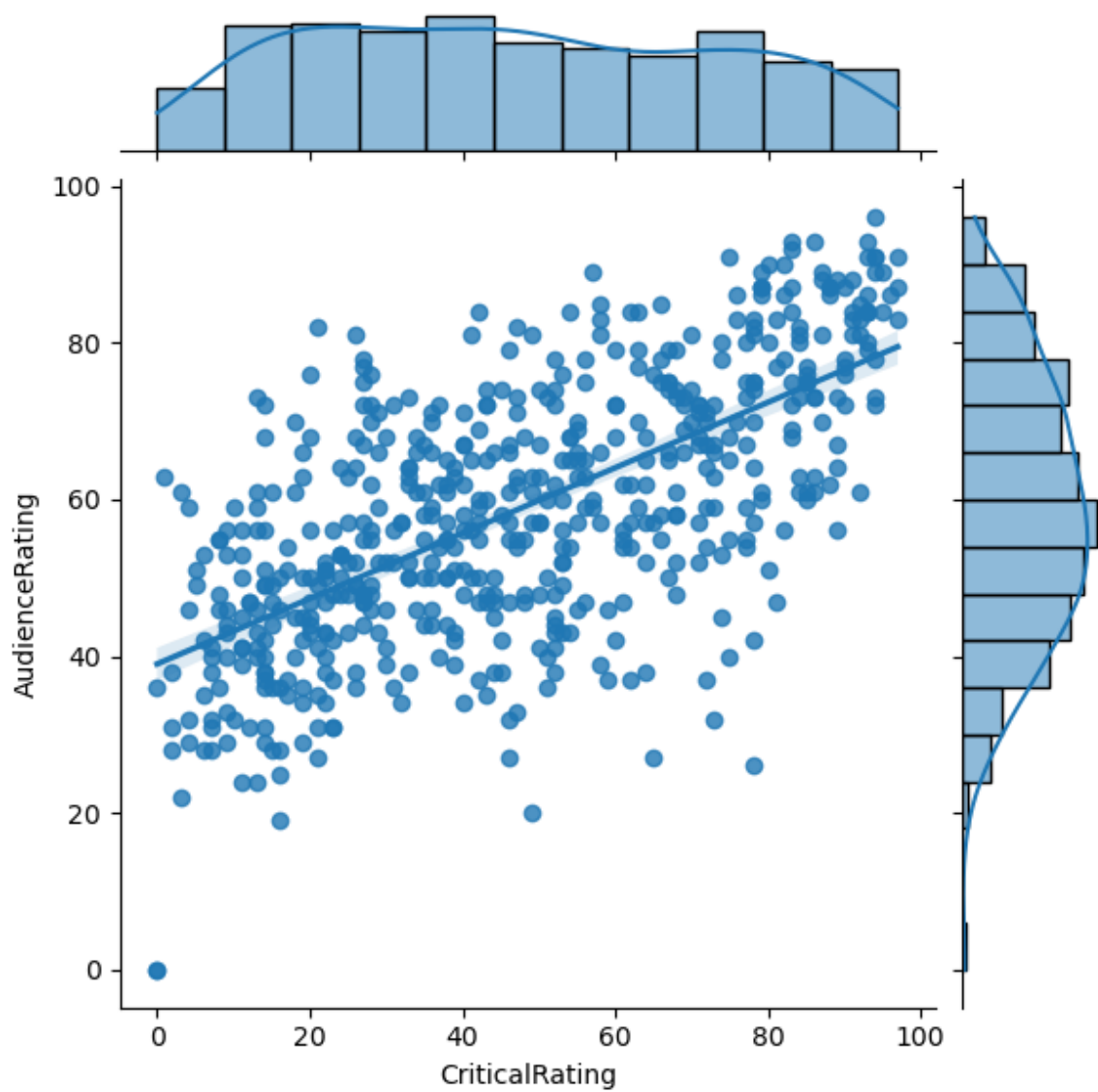
```
In [26]: j = sns.jointplot( data = movies , x ='CriticalRating',y = 'AudienceRating')
```



```
In [27]: j = sns.jointplot( data = movies , x ='CriticalRating',y = 'AudienceRating')
```

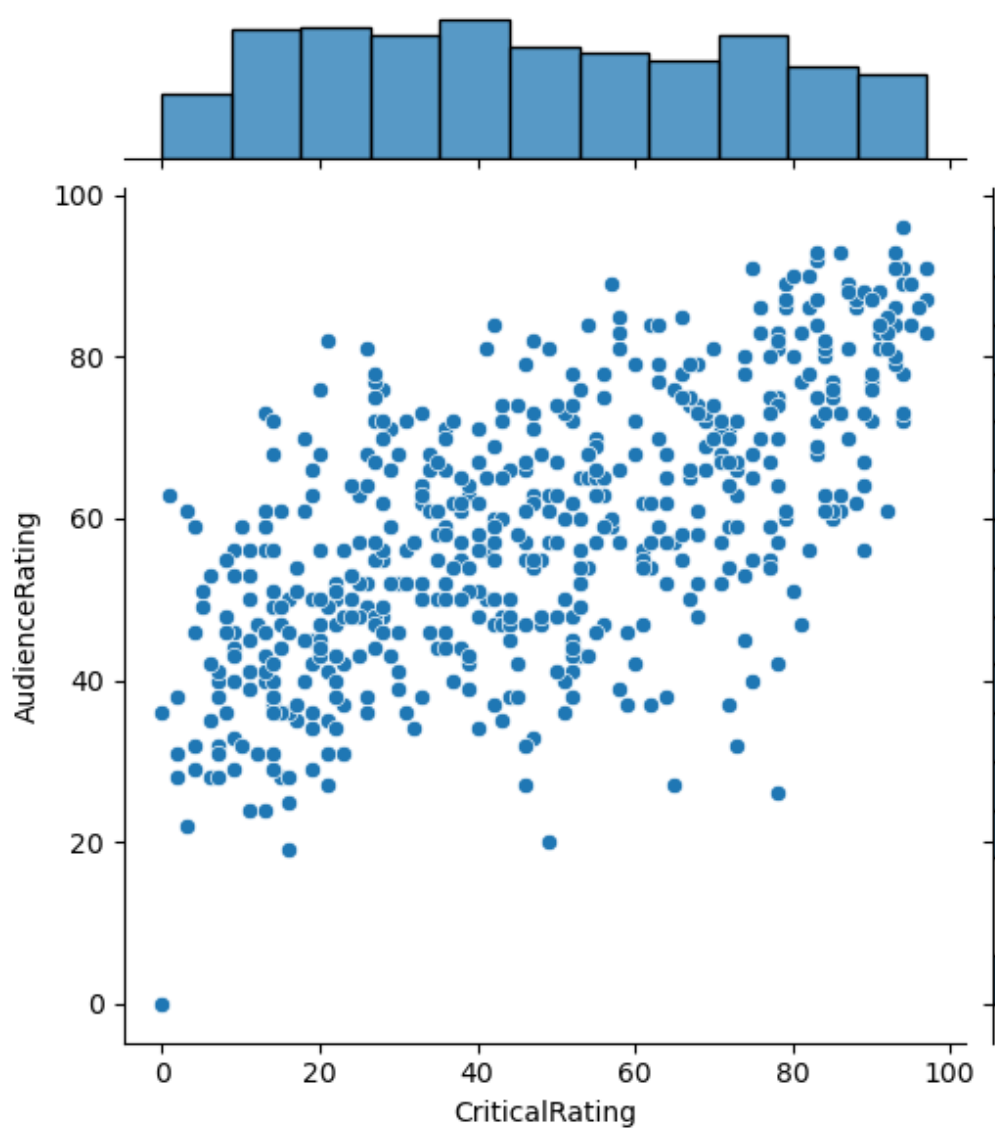


```
In [28]: j = sns.jointplot( data = movies , x ='CriticalRating',y = 'AudienceRating')
```



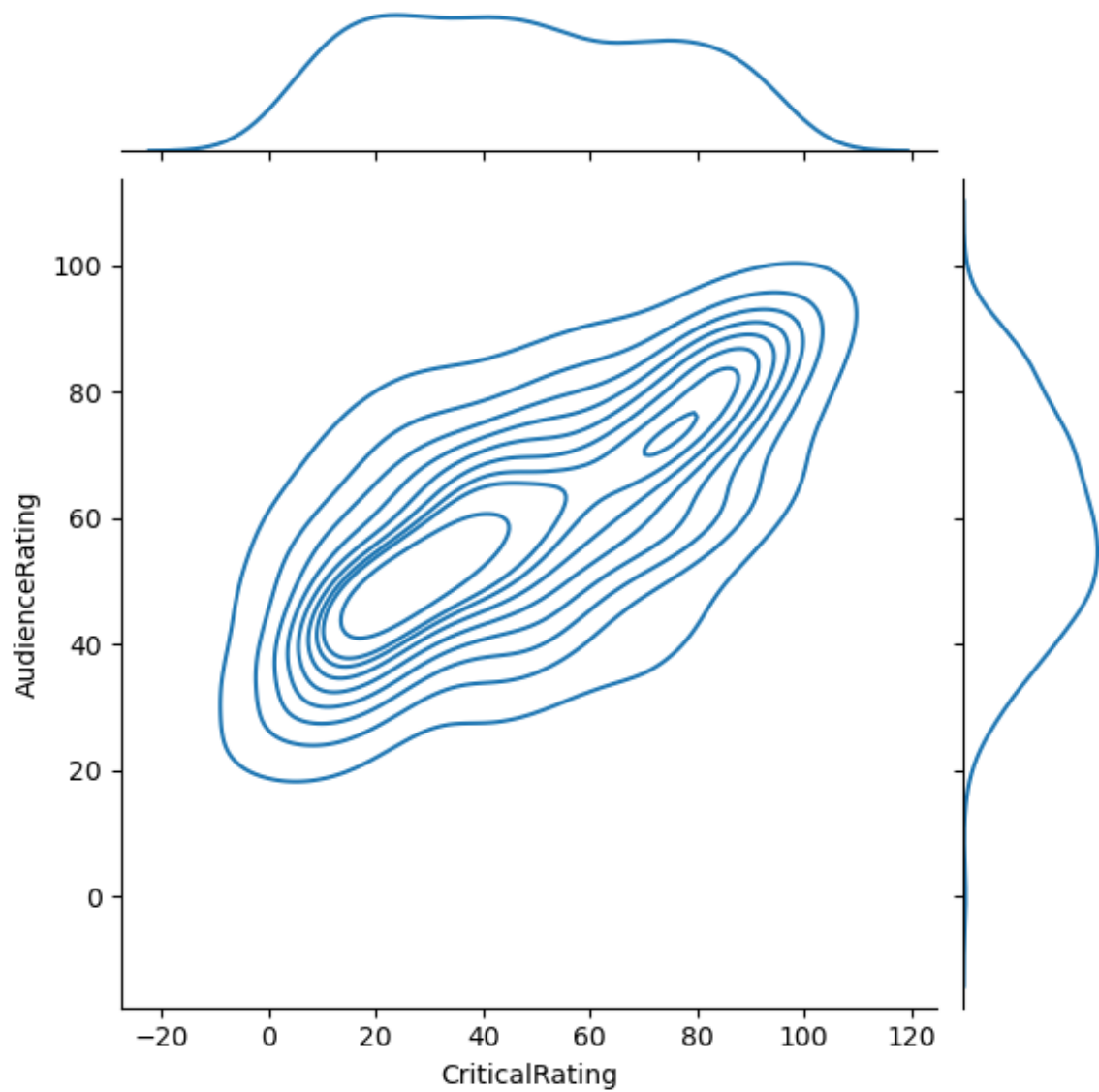
```
In [29]: sns.jointplot( data = movies , x ='CriticalRating',y = 'AudienceRat
```

```
Out[29]: <seaborn.axisgrid.JointGrid at 0x14e66ab10>
```



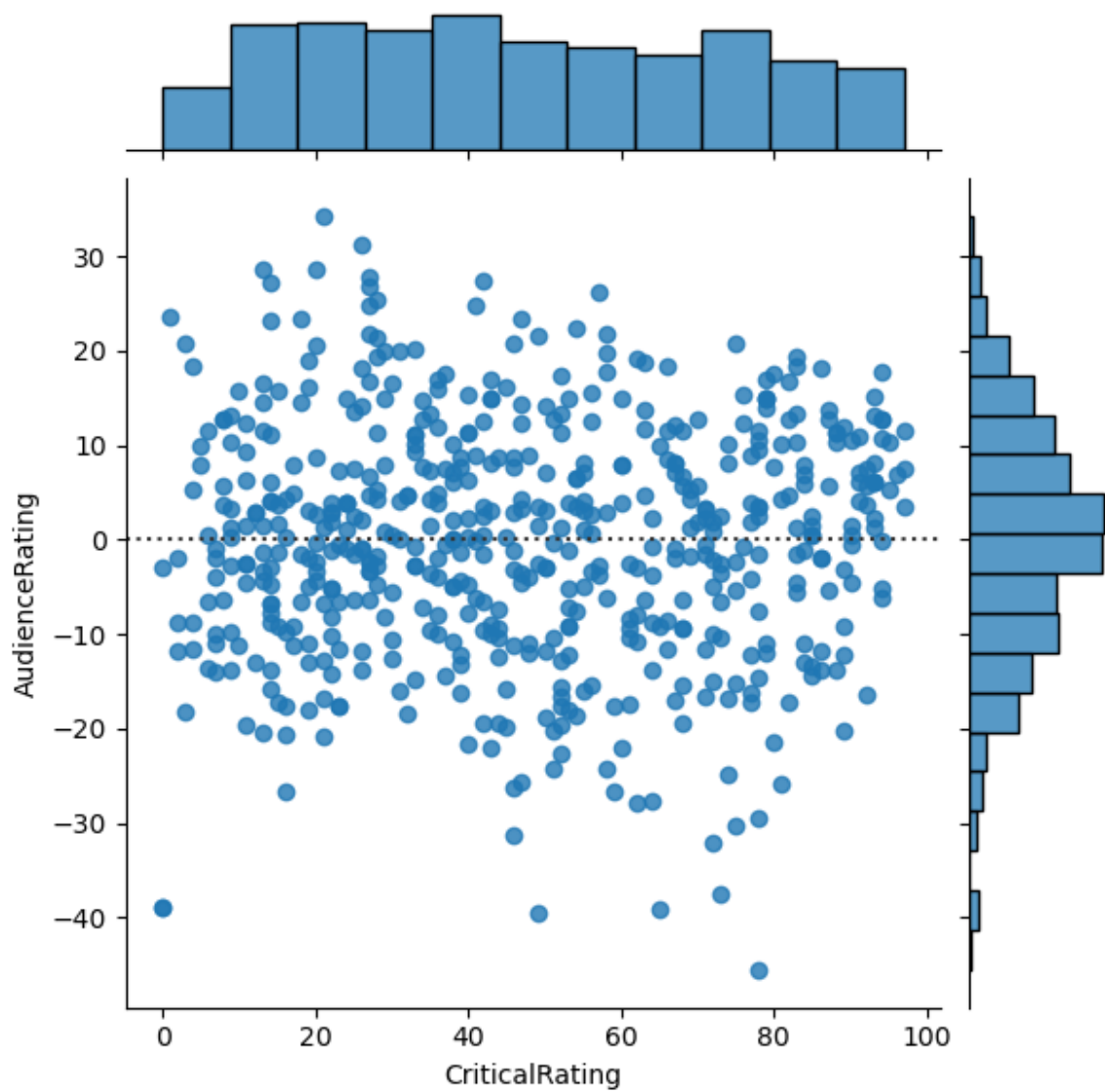
```
In [30]: sns.jointplot( data = movies , x ='CriticalRating',y = 'AudienceRating')
```

```
Out[30]: <seaborn.axisgrid.JointGrid at 0x14dda7bb0>
```



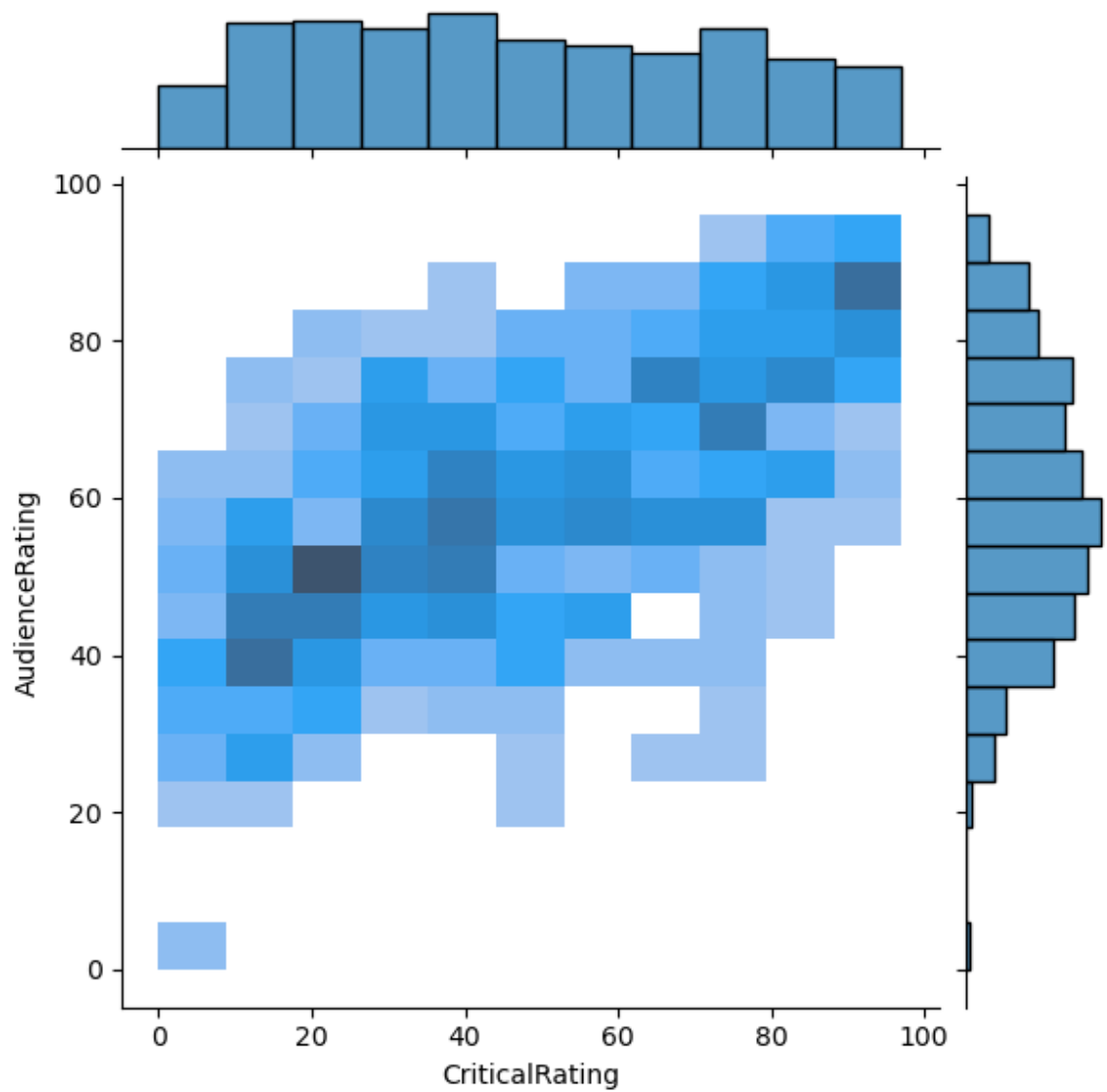
```
In [31]: sns.jointplot( data = movies , x ='CriticalRating',y = 'AudienceRat.
```

```
Out[31]: <seaborn.axisgrid.JointGrid at 0x14dde1fd0>
```

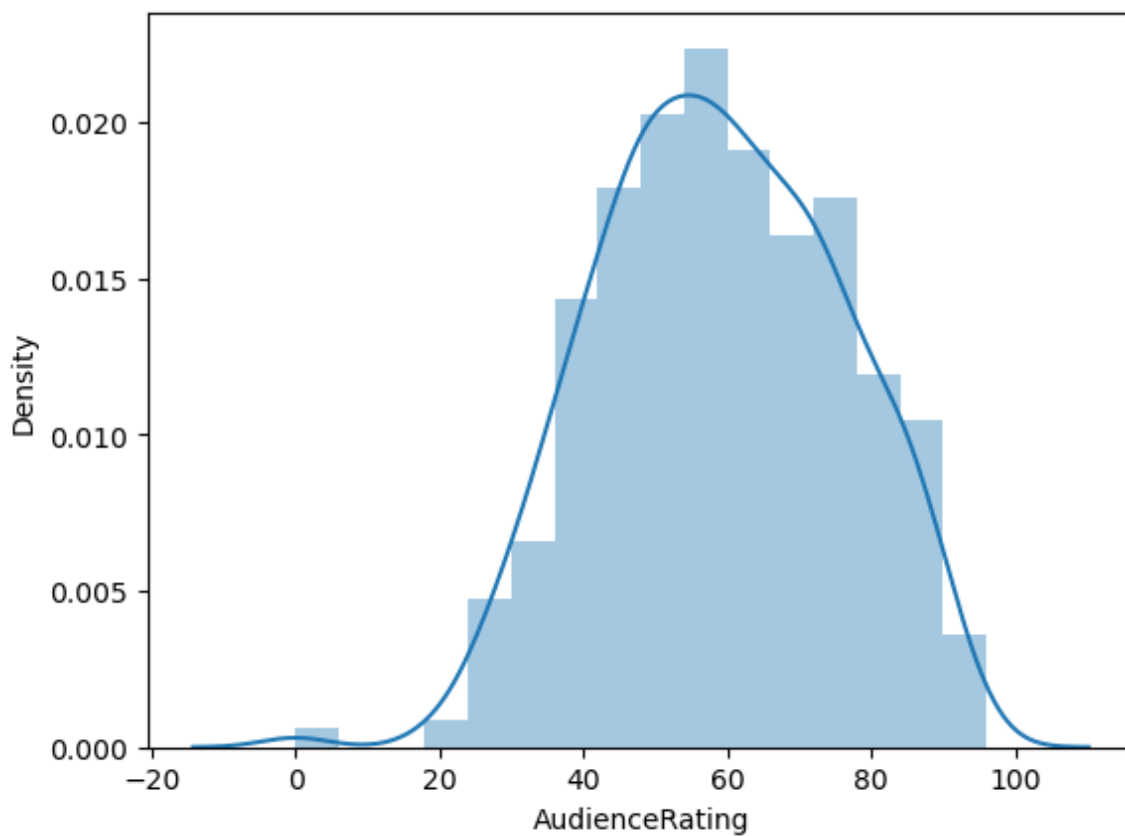


```
In [32]: sns.jointplot( data = movies , x ='CriticalRating',y = 'AudienceRat
```

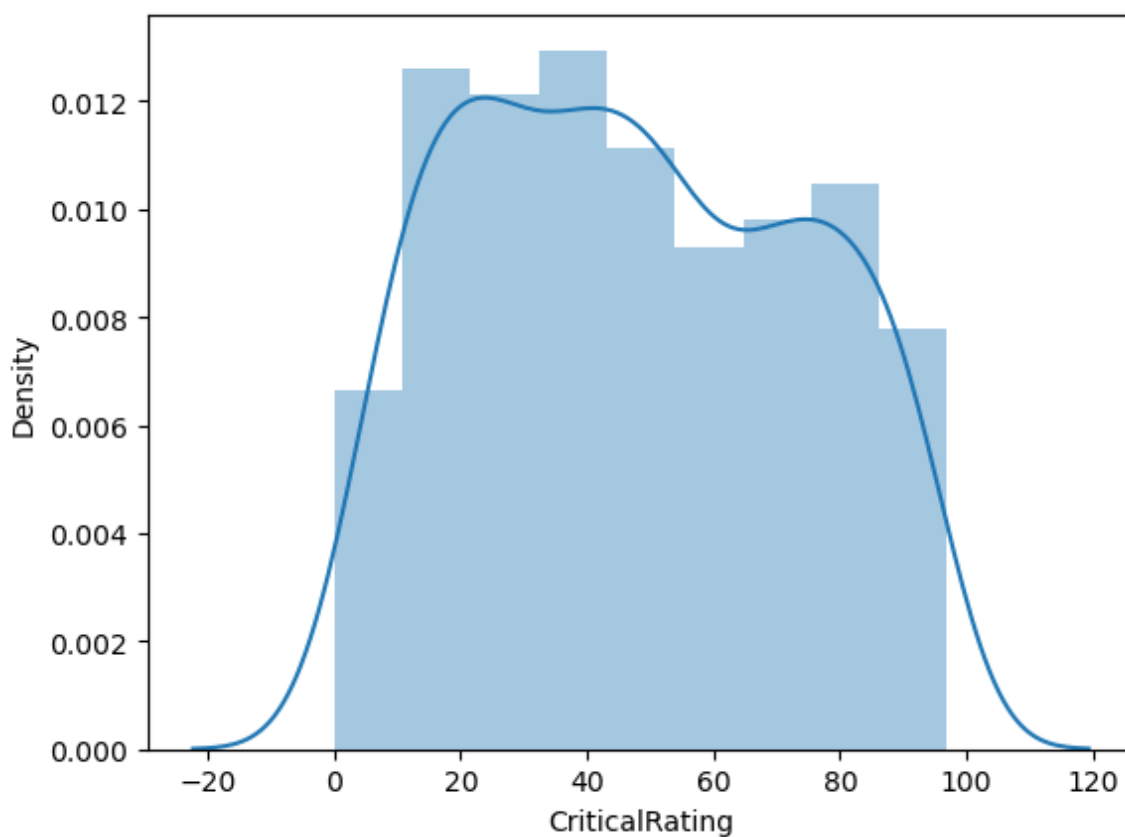
```
Out[32]: <seaborn.axisgrid.JointGrid at 0x14dd83240>
```



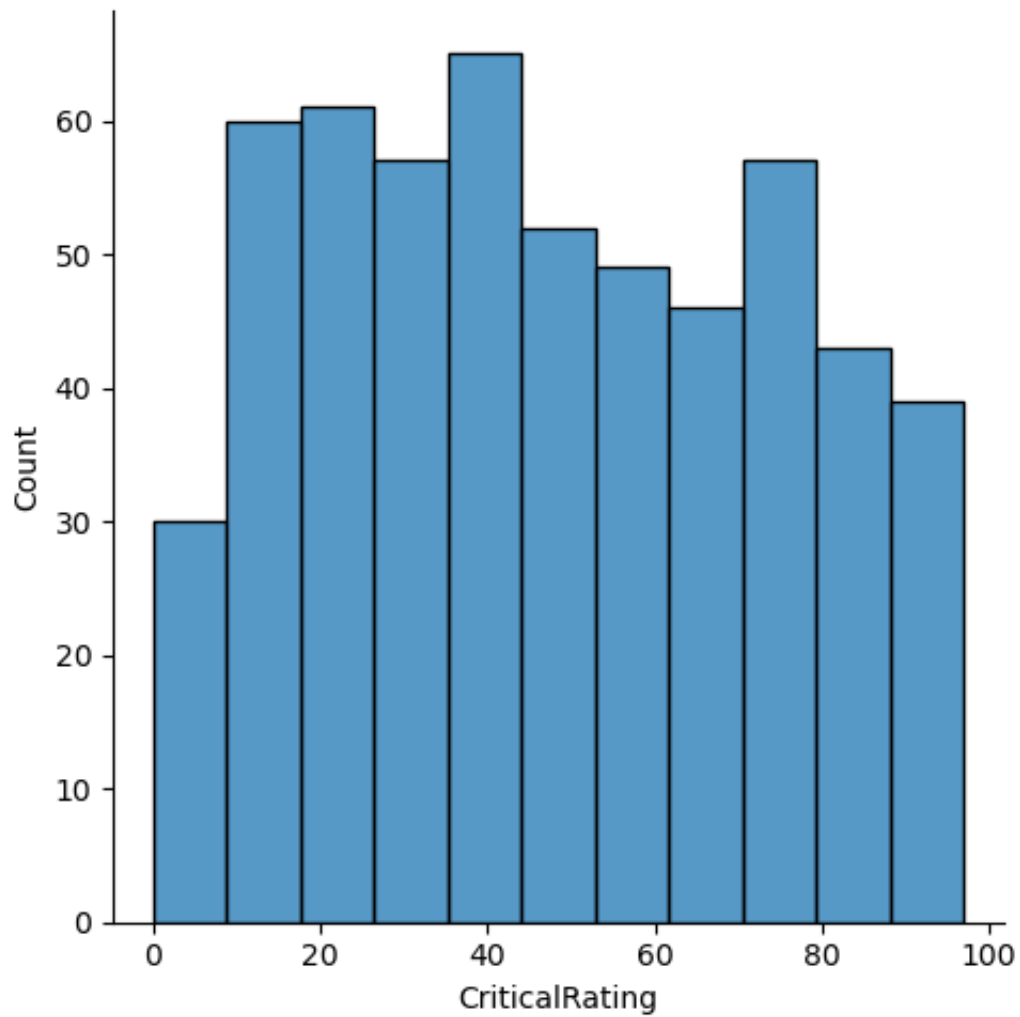
```
In [33]: m1 = sns.distplot(movies.AudienceRating)
```



```
In [34]: m1 = sns.distplot(movies.CriticalRating)
```

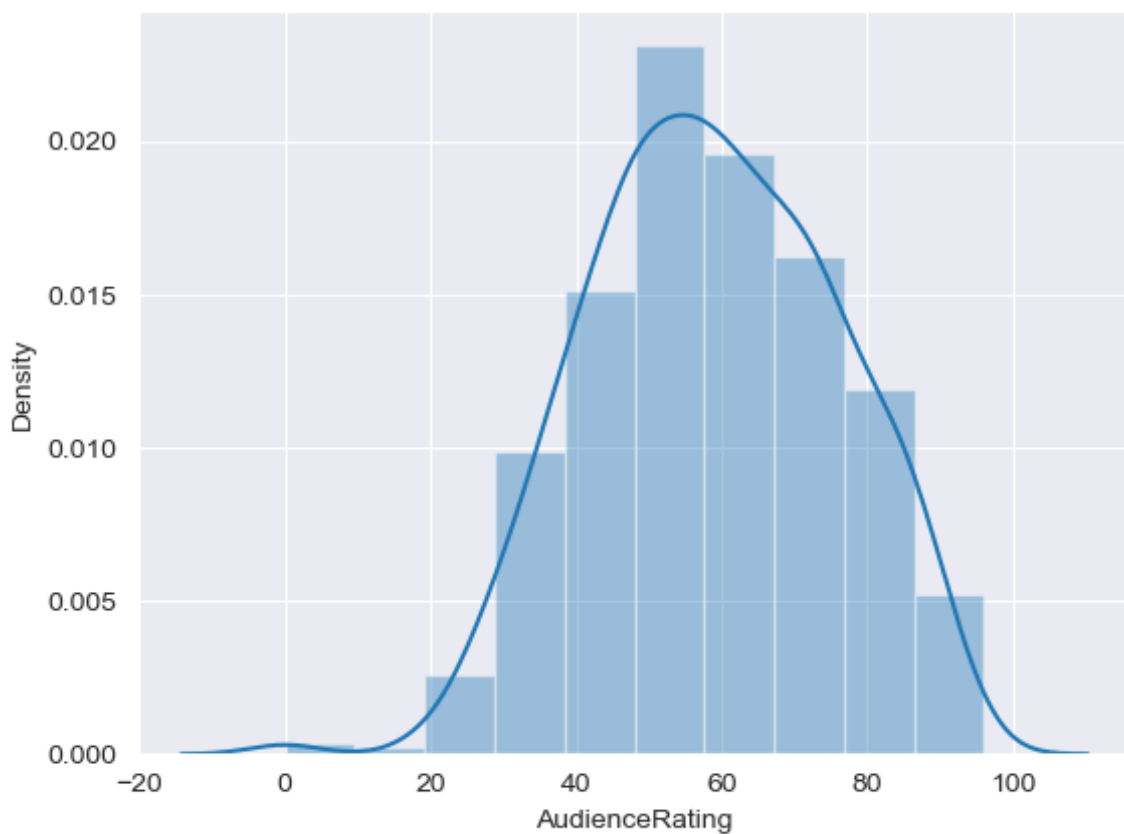


```
In [35]: m1 = sns.distplot(movies.CriticalRating)
```

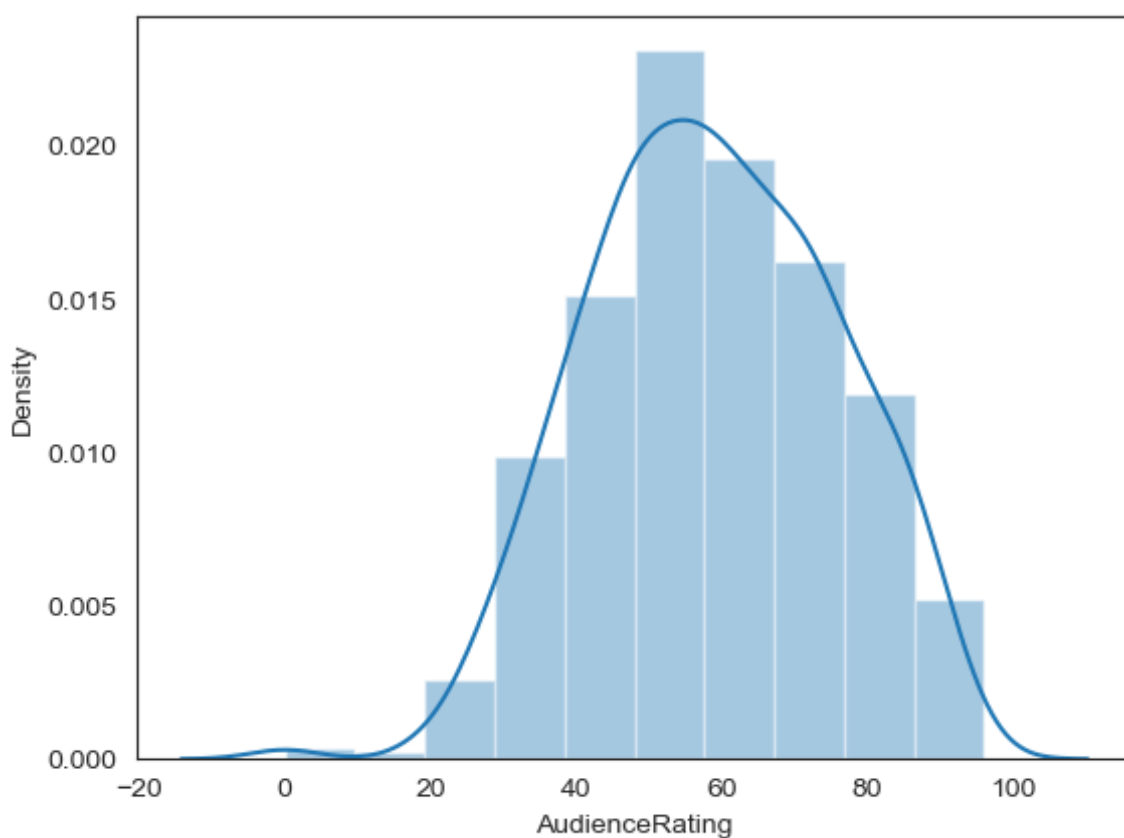


```
In [36]: sns.set_style('darkgrid')
```

```
In [37]: m2 = sns.distplot(movies.AudienceRating,bins = 10)
```



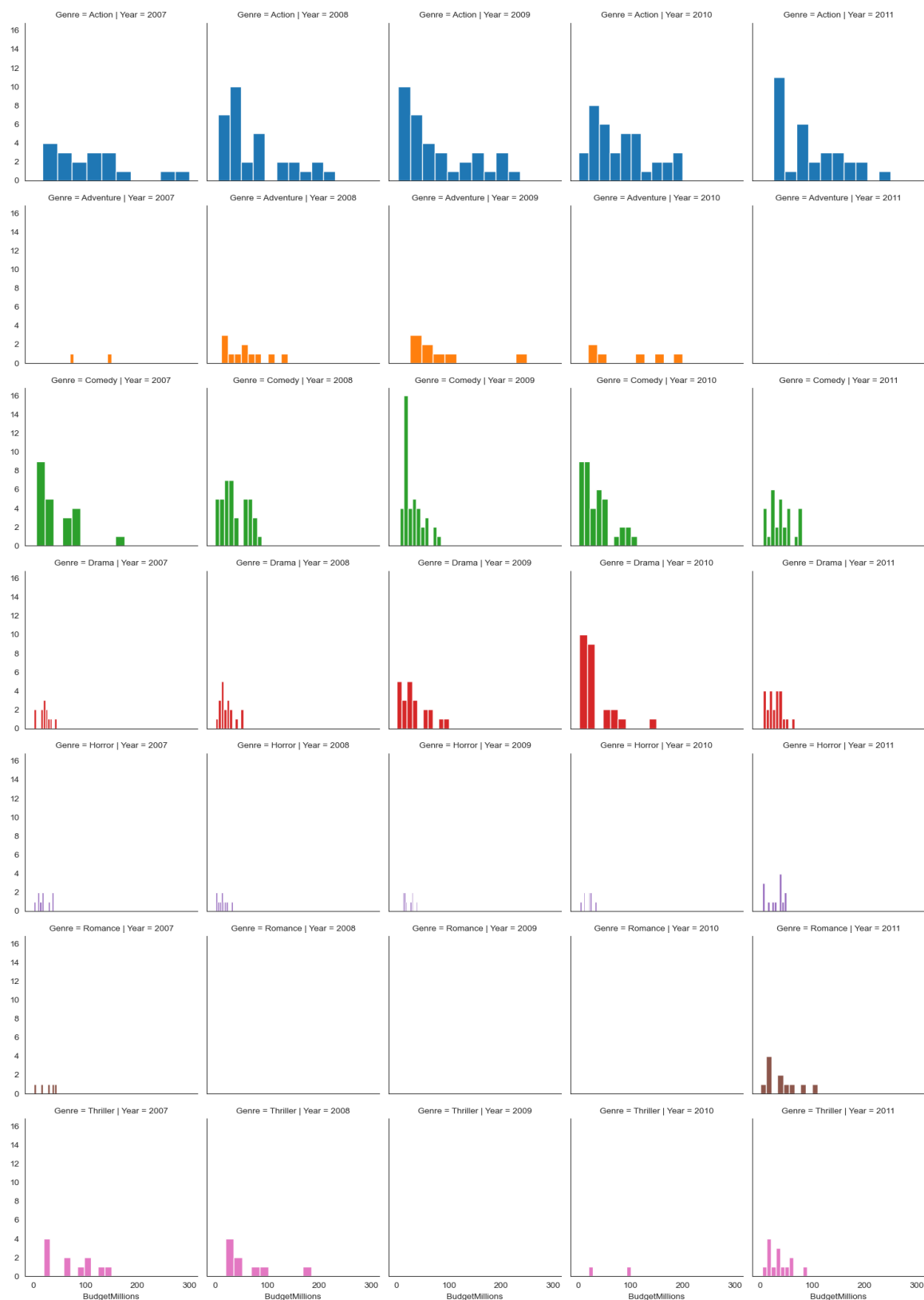
```
In [38]: sns.set_style('white')
m3 = sns.distplot(movies.AudienceRating, bins = 10)
```



```
In [39]: g = sns.FacetGrid (movies, row = 'Genre', col = 'Year', hue = 'Genre')
```



```
In [43]: g = sns.FacetGrid (movies, row = 'Genre', col = 'Year', hue = 'Genre')
g = g.map(plt.hist, 'BudgetMillions') #scatterplots are mapped in f
```



In [] :